2024 4th International Conference on Advancement in Electronics & Communication Engineering (AECE) | 979-8-3503-6472-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/AECE62803.2024.10911835

AgricAI Hub: Integrating Deep Learning Innovations to Empower Farmers

Shweta Jaiswal
Dept. of Physics Bilaspur University
Bilaspur,India
jheel.28@gmail.com

A.K Shrivastava
Dept. of Physics Raman University
Chhattisgarh, India
drakshrivastava01@gmail.com

Saurabh Pratap Singh Rathore
Dept. of Management, ICAPSR, New Delhi, India
rathoresaurabhsingh@gmail.com

Pratyksh Pandey
Dept. of Agriculture Mangalayatan University Aligarh, India
pratyksh.pandey@mangalayatan.edu.in

R.Subalakshmi
Dept. of CSE, Dhanalakshmi Srinivasan University Coimbatore, India
suba.dsravindran@gmail.com

Sourav Mondal
Dept. of management, Symbiosis International University Pune, India
sourav.mondal@scmspune.ac.in

Abstract—AgricAI Hub stands at the forefront of agricultural technology, offering a comprehensive platform designed to empower farmers through advanced deep learning capabilities. This integrated solution addresses various challenges encountered in agricultural operations. Using cuttingedge deep learning algorithms, AgricAI Hub excels in early detection of plant diseases with remarkable precision, achieving an accuracy rate of 97.94% across 20 different crops. This breakthrough enables swift identification and mitigation of crop threats, minimizing losses and optimizing yields. Additionally, AgricAI Hub supports farmers in optimizing agricultural practices through real time climate analysis, facilitating informed decisions on activities like harvesting and nutrient management. The platform includes a cropspecific fertilizer calculator that accurately determines nutrient requirements, promoting soil health and crop vitality. Multilingual support in Marathi, Hindi, Punjabi, and English ensures accessibility for diverse farming communities, enhancing adoption and impact. In summary, AgricAI Hub leverages deep learning to equip farmers with actionable insights and tools, driving innovation and sustainability in agriculture.

Keywords—AgricAI Hub, Agricultural aid, Deep learning technology, Plant disease detection, Real time climate analysis, Crop specific fertilizer calculator, Sustainable farming practices, Multilingual support

I. Introduction

Agriculture is a vital sector, especially in developing regions where it forms the core of rural economies and supports millions of livelihoods. However, significant issues like pest infestations, plant diseases, and unpredictable weather contribute to massive crop losses, intensifying food insecurity and economic instability. According to the Associated Chambers of Commerce and Industry of India, crop losses from pests and diseases alone can reach Rs. 50,000 crore ($500 billion) annually, posing severe risks to food security.

To address these challenges, this paper introduces Agricultural Aid, an Android application aimed at empowering farmers through machine learning technology. Agricultural Aid provides functionalities like plant disease detection, weather forecasting, fertilizer calculation, and multilingual support, addressing crucial agricultural needs. The app's plant disease detection feature uses machine learning algorithms that apply both image processing and deep learning techniques for accurate disease identification. The image processing approach leverages methods like color space conversion and thresholding, while the deep learning approach uses CNN for higher accuracy in disease classification.

Through machine learning, Agricultural Aid seeks to modernize farming practices, reduce crop losses, and support global food security, ultimately creating a more resilient and sustainable agricultural future.

II. Literature Review

Altalak et al. (2022) This study surveys the application of deep learning technologies in smart agriculture, focusing on how AIbased systems improve productivity and sustainability in farming [1]. The review explores various use cases and challenges in implementing these technologies in agricultural practices.Albahar (2023) Albahar discusses the impact of deep learning on agriculture, highlighting its transformative role in crop management, disease detection, and yield prediction [2]. The paper also addresses challenges such as data quality, computational requirements, and scalability.

Kaushik (2024) This research paper focuses on dynamic data scaling techniques for streaming machine learning. It discusses the challenges and solutions related to real time data processing and scaling in machine learning applications, with relevance to industries like agriculture that deal with large datasets [3]. Abdul Ameer et al. (2024) The paper explores the integration of IoT, UAVs, and deep learning in agriculture to empower farmers [4]. It discusses how these technologies facilitate precision farming, enhancing crop monitoring, disease detection, and resource management in agricultural systems. Rathor et al. (2024) Rathor and colleagues review the role of IoT and AIdriven technologies in vertical farming [5]. The paper highlights innovations that optimize resource usage, increase productivity, and reduce environmental impact, providing a comprehensive look at these emerging technologies in controlled agriculture systems.

Mana et al. (2024) Mana et al. examine the sustainable application of AI in agricultural production, focusing on how AI technologies can enhance farm management and improve sustainability [6]. They explore the potential implications of AIdriven agricultural practices for environmental and economic sustainability. Lu et al. (2017) This study presents an infield automatic wheat disease diagnosis system using AI technologies [7]. The research demonstrates how machine learning and computer vision can be employed to detect and classify wheat diseases in real time, improving crop health management and yield prediction.

Authorized licensed use limited to: Nitte Meenakshi Institute of Technology. Downloaded on August 13,2025 at 05:21:46 UTC from IEEE Xplore. Restrictions apply.

Fuentes et al. (2017) This study introduces a deep learning model for real time detection of diseases and pests in tomato plants, emphasizing the model's robustness and potential for largescale agricultural use through improved disease management [8]. Kerkech et al. (2018) The authors apply deep learning with colorimetric spaces and vegetation indices to detect vine diseases using UAV imagery, showcasing how this approach improves disease detection accuracy in vineyards [9]. Adhitya et al. (2023) This paper presents an IoT and deep learning based system to enhance farmer safety by monitoring environmental hazards, underscoring the role of AI in agricultural risk management [10].

Hu et al. (2019) Focusing on lowshot learning, this study proposes a model for identifying tea leaf diseases with limited data, highlighting AI's potential in situations with restricted training datasets [11]. Coulibaly et al. (2019) The authors leverage deep neural networks and transfer learning to identify millet crop diseases, demonstrating the effectiveness of AI in processing crop images for accurate disease identification [12]. Rathore & Rathore (2024) Although centered on human resources, this study explores machine learning applications in predicting employee turnover and performance, showcasing AI's versatility beyond agriculture [13]. Picon et al. (2019) This paper uses deep convolutional neural networks on mobilecaptured images to classify crop diseases, illustrating the practicality of mobilebased solutions for realworld disease diagnosis [14]. Rahnemoonfar & Sheppard (2017) This study presents Deep Count, a deep learning model for accurately counting fruits on trees using simulated data, demonstrating the application of deep learning in estimating fruit yield [15].

Veeramani et al. (2018) The authors propose DeepSort, a convolutional neural networkbased approach for sorting haploid maize seeds, offering a precise method for seed selection in agricultural practices [16]. Kaushik et al. (2024) Although centered on transportation, this paper discusses security and privacy in intelligent systems, with potential applications in secure data handling for agricultural IoT and AI systems [17]. Altuntaş et al. (2019) This paper uses CNNs with transfer learning to differentiate between haploid and diploid maize seeds, providing insights into the automation of seed sorting in agriculture [18].

In summary, AI and deep learning are revolutionizing agriculture by enhancing efficiency and sustainability through precision farming. While promising, challenges like data scalability and environmental impacts require further research to support resilient, adaptable agricultural systems.

III. Materials And Methods

The work was conducted in two primary phases: initially, 100 prior research studies exploring the intersection of deep learning (DL) and agriculture were gathered. Following this, a comprehensive evaluation and analysis of the compiled literature were undertaken. This process involved scrutinizing the methodologies, findings, and implications of each study to glean insights into the role of DL in agricultural contexts. Additionally, various aspects such as the efficacy of DL tech niques, their practical applications in agriculture, and potential challenges were thoroughly examined to provide a holistic understanding of the subject matter.

Types of Data Gathered:

- Remote Sensing Data

Remote sensing data is essential for agricultural monitoring, offering valuable insights for crop health, land use, and environmental management. It aids AgricAI Hub by providing accurate, spatially explicit information for informed decisionmaking. Key applications include:

1. Crop Monitoring and Management: Satellite and aerial imagery allow early detection of crop stress, disease, and nutrient deficiencies, supporting targeted interventions for improved crop health and yield.

2. Land Use and Crop Mapping: Remote sensing enables crop classification and field boundary delineation, helping optimize land use and supporting precision agriculture.

3. Disease and Pest Detection: By identifying anomalies in vegetation reflectance, remote sensing helps AgricAI Hub detect disease and pest infestations early, enabling timely control measures.

4. Environmental Monitoring: Remote sensing assesses soil moisture, water bodies, and soil properties, informing water management and soil fertility recommendations to improve crop resilience.

5. Weather Forecasting and Climate Resilience: Remote sensing contributes to climate data analysis, providing localized weather forecasts and climate risk assessments to support adaptive management strategies.

Integrating remote sensing into AgricAI Hub empowers farmers with actionable insights, enhancing resource allocation, productivity, and sustainability. Through deep learning and data analysis, AgricAI Hub tailors solutions to address farmers' specific challenges, promoting better outcomes in crop production and management.

- IoT Sensor Data

IoT sensor data in agriculture enables real time monitoring of soil, weather, crop health, and environmental conditions. This data supports precision agriculture, allowing farmers to manage irrigation, monitor crop health, detect pests, and optimize nutrient and water management. Through AgricAI Hub, IoT data integrates with satellite imagery, weather forecasts, and agronomic models to provide actionable insights and predictive analytics. Farmers can remotely control systems for irrigation and climate management, enhancing productivity and sustainability. This IoT driven approach allows data driven decision making and adaptive strategies, empowering farmers to optimize resources and improve outcomes in crop production and farm management.

- Historical Data

1. Plant Village Dataset: Contains 70,000 highquality images of diseased and healthy plant leaves from nine species, supporting plant disease detection and classification. Images are split into training, testing, and validation sets, aiding researchers in plant health improvement.

2. Indian Weather Data: A comprehensive dataset from Kaggle, covering weather observations across India, including temperature, humidity, rainfall, wind speed, and pressure over a long period. It offers insights into climate patterns, regional variations, and supports applications in climate analysis and disaster preparedness.

3. Crop Recommendation Dataset: Aids precision agriculture by recommending suitable crops for specific farms

Authorized licensed use limited to: Nitte Meenakshi Institute of Technology. Downloaded on August 13,2025 at 05:21:46 UTC from IEEE Xplore. Restrictions apply.

based on nitrogen, phosphorous, and potassium levels, temperature, humidity, soil pH, and rainfall. The data helps farmers make informed decisions for optimal crop yield.

4. Agricultural Sensors for Irrigation: Includes data from soil moisture and DHT11 sensors that monitor moisture, temperature, and humidity. The data helps automate irrigation systems by adjusting water delivery based on soil moisture, minimizing waste.

1. Historical agricultural datasets like the Plant Village Dataset, Indian Weather Data, Crop Recommendation Dataset, and Agricultural Sensors for Irrigation contribute significantly to modern farming practices. The Plant Village Dataset provides diverse images for machine learning in plant disease detection, supporting crop health improvement. The Indian Weather Data offers extensive coverage of climate variables, aiding weather trend analysis and applications in agriculture and disaster management. The Crop Recommendation Dataset promotes precision farming by suggesting crops based on soil and climate parameters, enhancing productivity. Lastly, agricultural sensors support automated irrigation by controlling water supply based on soil moisture, ensuring efficiency and resource conservation. Together, these datasets enhance datadriven agriculture and sustainability.

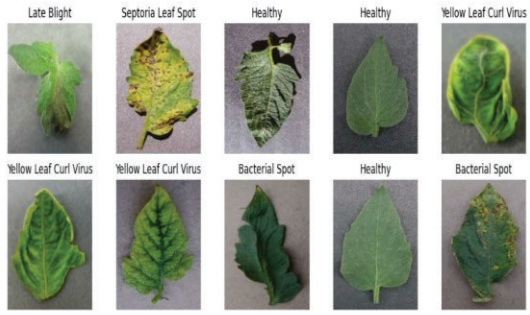

Fig. 1. Sample Image of "Tomato" dataset.

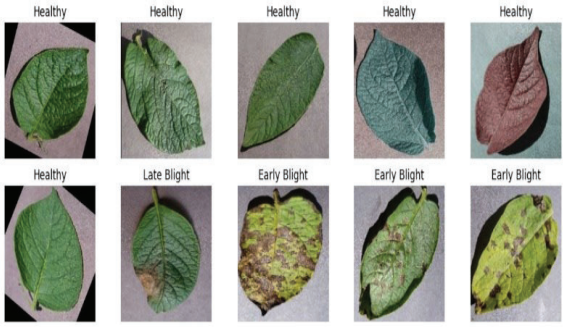

Fig. 2. Sample Image of "Apple" dataset

Categories: Agricultural Science, Machine Learning, Internet of Things, Precision Agriculture, Irrigation Management, Soil Moisture, Humidity, and Air Temperature.

Techniques and algorithms used in AgricAI Hub: In the AgricAI Hub project, various deep learning techniques and algorithms can be applied to analyze agricultural data and provide valuable insights to farmers. In this project we have used some types of deep learning techniques and algorithms that can be used in this project:

1)Convolutional Neural Networks (CNNs)

2) Recurrent Neural Networks (RNNs)

3) Generative Adversarial Networks (GANs)

4) Long Short-Term Memory Networks (LSTMs)

5) Deep Reinforcement Learning (DRL)

6) Transfer Learning

7) Ensemble Learning

Variable	Measurement Unit	Typical Range
Leaf Area	Square centimeters (cm^2) or square millimeters (mm^2)	Varies based on plant species and leaf size
Leaf Length	Centimeters (cm) or millimeters (mm)	Varies based on plant species and leaf size
Leaf Width	Centimeters (cm) or millimeters (mm)	Varies based on plant species and leaf size
Leaf Color (RGB)	None (color values)	0 to 255 (for each RGB component)
Disease Severity Score	None (e.g., 0 to 100, or 1 to 5)	Varies based on scoring system used
Leaf Texture	None (qualitative descriptor)	N/A
Plant Height	Meters (m) or centimeters (cm)	Varies based on plant species and growth stage
Plant Yield	Kilograms (kg) or grams (g)	Varies based on crop type and harvest yield
Plant Growth Stage	None (categorical variable)	Varies based on plant species and growth cycle

Fig. 3. Sample Image of "Grape" dataset

last_updated_epoc	sunrise	sunset	moonrise	moonset	moon_phase	moon_illun
1693000001	5:59 AM	6:41 PM	5:42 PM	3:38 AM	Waxing Gibbous	93
1693000002	6:00 AM	6:40 PM	5:39 PM	3:41 AM	Waxing Gibbous	93
1693000003	5:56 AM	6:34 PM	5:32 PM	3:39 AM	Waxing Gibbous	93
1693000004	6:00 AM	6:38 PM	5:36 PM	3:43 AM	Waxing Gibbous	93
1693000005	6:01 AM	6:39 PM	5:38 PM	3:42 AM	Waxing Gibbous	93
1693000006	6:03 AM	6:42 PM	5:42 PM	3:44 AM	Waxing Gibbous	93
1693000007	5:51 AM	6:31 PM	5:30 PM	3:32 AM	Waxing Gibbous	93
1693000008	5:55 AM	6:34 PM	5:33 PM	3:36 AM	Waxing Gibbous	93
1693000009	5:49 AM	6:31 PM	5:32 PM	3:27 AM	Waxing Gibbous	93
1693000010	6:08 AM	6:48 PM	5:47 PM	3:49 AM	Waxing Gibbous	93
1693000011	5:45 AM	6:26 PM	5:27 PM	3:23 AM	Waxing Gibbous	93
1693000012	5:47 AM	6:26 PM	5:25 PM	3:28 AM	Waxing Gibbous	93
1693000013	5:51 AM	6:28 PM	5:26 PM	3:34 AM	Waxing Gibbous	93
1693000014	6:12 AM	6:50 PM	5:49 PM	3:55 AM	Waxing Gibbous	93
1693000015	5:47 AM	6:28 PM	5:29 PM	3:25 AM	Waxing Gibbous	93
1693000016	5:51 AM	6:34 PM	5:35 PM	3:29 AM	Waxing Gibbous	93
1693000017	6:08 AM	6:47 PM	5:46 PM	3:50 AM	Waxing Gibbous	93
1693000018	6:10 AM	6:51 PM	5:50 PM	3:52 AM	Waxing Gibbous	93
1693000019	6:03 AM	6:42 PM	5:40 PM	3:46 AM	Waxing Gibbous	93
1693000020	5:56 AM	6:36 PM	5:36 PM	3:36 AM	Waxing Gibbous	93
1693000021	6:10 AM	6:52 PM	5:53 PM	3:50 AM	Waxing Gibbous	93
1693000022	5:55 AM	6:37 PM	5:38 PM	3:33 AM	Waxing Gibbous	93
1693000023	6:01 AM	6:43 PM	5:44 PM	3:40 AM	Waxing Gibbous	93
1693000024	6:07 AM	6:46 PM	5:46 PM	3:49 AM	Waxing Gibbous	93
1693000025	6:10 AM	6:51 PM	5:52 PM	3:50 AM	Waxing Gibbous	93
1693000026	6:10 AM	6:47 PM	5:45 PM	3:53 AM	Waxing Gibbous	93
1693000027	6:02 AM	6:46 PM	5:48 PM	3:40 AM	Waxing Gibbous	93
1693000028	5:56 AM	6:41 PM	5:45 PM	3:32 AM	Waxing Gibbous	93

Fig. 4. Sample dataset Air quality information

last_updated_epoch	country	location_name	region	latitude	longitude	timezone	last_updated
1693000001	India	Ashoknagar	Madhya Pradesh	24.57	77.72	Asia/Kolkat	8/29/23 10:45
1693000002	India	Raisen	Madhya Pradesh	23.33	77.8	Asia/Kolkat	8/29/23 10:45
1693000003	India	Chhindwara	Madhya Pradesh	22.07	78.93	Asia/Kolkat	8/29/23 10:45
1693000004	India	Betul	Madhya Pradesh	21.86	77.93	Asia/Kolkat	8/29/23 10:45
1693000005	India	Hoshangabad	Madhya Pradesh	22.75	77.72	Asia/Kolkat	8/29/23 10:45
1693000006	India	Sehore	Madhya Pradesh	23.2	77.08	Asia/Kolkat	8/29/23 10:45
1693000007	India	Jabalpur	Madhya Pradesh	23.17	79.95	Asia/Kolkat	8/29/23 10:45
1693000008	India	Narsimhapur	Madhya Pradesh	22.95	79.2	Asia/Kolkat	8/29/23 10:45
1693000009	India	Panna	Madhya Pradesh	24.72	80.2	Asia/Kolkat	8/29/23 10:45
1693000010	India	Ujjain	Madhya Pradesh	23.18	75.77	Asia/Kolkat	8/29/23 10:45
1693000011	India	Rewa	Madhya Pradesh	24.53	81.3	Asia/Kolkat	8/29/23 10:45
1693000012	India	Dindori	Madhya Pradesh	22.95	81.08	Asia/Kolkat	8/29/23 10:45
1693000013	India	Balaghat	Madhya Pradesh	21.78	80.25	Asia/Kolkat	8/29/23 10:45
1693000014	India	Barwani	Madhya Pradesh	22.03	74.9	Asia/Kolkat	8/29/23 10:45
1693000015	India	Satna	Madhya Pradesh	24.58	80.83	Asia/Kolkat	8/29/23 10:45
1693000016	India	Chhatarpur	Madhya Pradesh	24.9	79.6	Asia/Kolkat	8/29/23 10:45
1693000017	India	Indore	Madhya Pradesh	22.72	75.83	Asia/Kolkat	8/29/23 10:45
1693000018	India	Ratlam	Madhya Pradesh	23.32	75.07	Asia/Kolkat	8/29/23 10:45
1693000019	India	Harda	Madhya Pradesh	22.33	77.1	Asia/Kolkat	8/29/23 10:45
1693000020	India	Sagar	Madhya Pradesh	23.83	78.72	Asia/Kolkat	8/29/23 10:45
1693000021	India	Neemuch	Madhya Pradesh	24.48	74.87	Asia/Kolkat	8/29/23 10:45
1693000022	India	Tikamgarh	Uttar Pradesh	24.73	78.83	Asia/Kolkat	8/29/23 10:45
1693000023	India	Guna	Madhya Pradesh	24.65	77.32	Asia/Kolkat	8/29/23 10:45
1693000024	India	Dewas	Madhya Pradesh	22.97	76.07	Asia/Kolkat	8/29/23 10:45
1693000025	India	Mandsaur	Madhya Pradesh	24.07	75.07	Asia/Kolkat	8/29/23 10:45
1693000026	India	Khargone	Madhya Pradesh	21.82	75.6	Asia/Kolkat	8/29/23 10:45
1693000027	India	Sheopur	Madhya Pradesh	25.67	76.7	Asia/Kolkat	8/29/23 10:45
1693000028	India	Morena	Madhya Pradesh	26.5	78	Asia/Kolkat	8/29/23 10:45

Fig. 5. Sample dataset Location information

The bibliographic analysis in the domain based on databases of the Science Citation Index (SCI) included full text papers published in peer reviewed journals. A keyword based search for these papers was done by using search terms, "deep learning", and "agriculture or farming". Through that query, 133 papers were obtained. Articles containing the keywords "deep learning" but not related to the agricultural area have been eliminated. As a result of this search, 133 articles were identified, initially. Some articles were excluded due to the lack of meaningful findings and the initial number

Authorized licensed use limited to: Nitte Meenakshi Institute of Technology. Downloaded on August 13,2025 at 05:21:46 UTC from IEEE Xplore. Restrictions apply.

of papers was reduced to 130. Then review papers were excluded from the scope of the study, and the final number of papers was 120.

After the collection of related work, a detailed review, and analysis of this work were undertaken. Considering the following research questions, the 40 papers selected were analyzed individually:

Q1: What are the topics, where deep learning is implemented in the agriculture domain?

Q2: What are the problems they addressed?

Q3: What approaches were employed to solve the problems? Q4: What are the sources of data used?

Q5: What is the benefit of deep learning relative to other solutions?

IV. RESULT

In the Appendices, a list of the 20 reviewed works is given, indicating the research domain, the problem the research addresses, proposed methods for the solution, and sources of data used. The summary tables obtained from analyzes are given in Appendix A. The highest number of deep learning-based agriculture relevant papers on the database of the SCI appeared in 2019 (76) and there were no papers before 2016. The time trend analysis given in Table 1, displays the eight most productive countries. With its rapidly growing publications in 2019, China was a leader throughout the period. Similarly, the growth rate of publications in the USA was much faster than in the other six predominant countries. As shown in Table 1, the distribution of the topics focused on in all countries is almost equal, except papers by Chinese authors that concentrate on topics such as disease detection (6), land cover identification (6), object recognition (6), pest recognition (5), plant classification (4), and precision livestock farming (8). As shown in Table 2, those topics appear to be the most common. Therefore, it seems that China determines the trend of deep learning based agriculturally relevant studies.

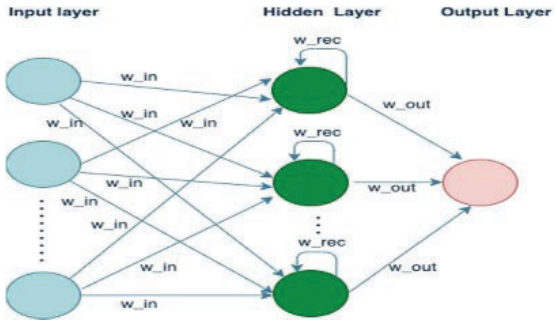

Fig. 6. A common CNN architecture

$$h_t = \sigma\left(W_{xh}^T x_t + W_{hh}^T h_{t-1} + b_h\right) \quad (1)$$

$$y_t = \sigma\left(W_0^T h_t + b_0\right) \quad (2)$$

Fig. 7. Simple RNN Model architecture.

A. Disease Detection

Plant disease detection is crucial to minimize production losses in agriculture. Various methods, tailored to specific plants, help monitor and control disease spread. Examples include Lu et al.'s automatic wheat disease diagnosis and Fuentes et al.'s tomato disease and pest detection, showcasing diverse, plant specific approaches detailed in literature.

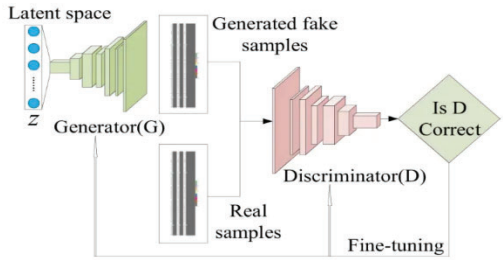

Fig. 8. Generative Adversarial Networks (GANs)

B. Plant Classification

Manual harvesting is labor intensive, prompting research into automated, crop specific systems. Grinblat et al. used vein morphology for plant identification, while Rahnemoon far and Sheppard automated yield estimation in tomatoes. CNN based methods are used by Veeramani et al. for sorting maize seeds. Further research on plant classification is in Appendix B.

C. Precision Livestock Farming

Precision livestock farming addresses animal health through monitoring indicators like comfort and behavior. Machinelearning techniques predict piglet skin temperature (Gorczyca et al.) and estimate cattle pose (Li et al.). These deep learning methods offer noninvasive, real time livestock management solutions, with more studies listed in Appendix B.

D. Object Recognition

Anomaly detection automates precision farming by identifying unexpected items or behaviors, essential for recognizing unnatural elements in agriculture. Christiansen et al. used deep learning with field homogeneity, while Ma et al. employed unsupervised hyperspectral anomaly detection. Zhang et al. developed AMTNet, effectively recognizing agricultural machinery via preprocessing of terminal camera images.

E. Pest Recognition

Identifying agricultural pests is crucial to reducing crop damage. Cheng et al. utilized deep residual learning for pest detection, while Patel et al. developed an AI vision system for pests like the Asian citrus psyllid. Various researchers, including Ding and Taylor, used CNN and transfer learning techniques to classify pests accurately.

F. Smart Irrigation

Efficient water usage in agriculture is essential given declining global resources. AlZu'bi et al. integrated IoT sensors and machine learning for water management, while Song et al. combined deep networks with cellular automata to predict soil moisture. Zambrano et al. used rainfall and climate data to forecast drought impacts on agriculture in Chile.

G. Phenotyping

Automated plant phenotyping reduces labor costs in crop improvement. Uzal et al. used deep learning to count seeds in soybeans, and Ampatzidis et al. applied UAVs for rapid data collection. Santos Ferreira et al. utilized CNNs to detect weeds in soybean fields, optimizing weed management and enhancing crop sustainability.

Authorized licensed use limited to: Nitte Meenakshi Institute of Technology. Downloaded on August 13,2025 at 05:21:46 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] M. Altalak, M. Ammad uddin, A. Alajmi, and A. Rizg, "Smart Agri culture Applications Using Deep Learning Technologies: A Survey," Applied Sciences, vol. 12, no. 12. MDPI AG, p. 5919, Jun. 10, 2022. doi: 10. 3390/app12125919.

[2] M. Albahar, "A Survey on Deep Learning and Its Impact on Agriculture: Challenges and Opportunities," Agriculture, vol. 13, no. 3. MDPI AG, p. 540, Feb. 23, 2023. doi: 10. 3390/agriculture13030540.

[3] Dr. P. Kaushik, "Dynamic Data Scaling Techniques for Streaming Machine Learning," International Journal for Global Academic & Sci entific Research, vol. 3, no. 1. International Consortium of Academic Professionals for Scientific Research, pp. 1–12, Apr. 04, 2024. doi: 10.55938/ijgasr. v3i1. 68.

[4] S. Abdul Ameer, M. A. Alkhafaji, Z. Jaffer, and M. AlFarouni, "Empowering Farmers with IoT, UAVs, and Deep Learning in Smart Agriculture," E3S Web of Conferences, vol. 491. EDP Sciences, p.04007, 2024. doi: 10. 1051/e3sconf/202449104007.

[5] A. S. Rathor, S. Choudhury, A. Sharma, P. Nautiyal, and G. Shah, "Empowering vertical farming through IoT and AIDriven technologies: A comprehensive review," Heliyon, vol. 10, no. 15. Elsevier BV, p. e34998, Aug. 2024. doi: 10. 1016/j. heliyon. 2024. e34998.

[6] A. A. Mana, A. Allouhi, A. Hamrani, S. Rehman, I. el Jamaoui, and K. Jayachandran, "Sustainable AIbased production agriculture: Exploring AI applications and implications in agricultural practices," Smart Agricultural Technology, vol. 7. Elsevier BV, p. 100416, Mar.2024. doi: 10. 1016/j. atech. 2024. 100416.

[7] J. Lu, J. Hu, G. Zhao, F. Mei, and C. Zhang, "An infield automatic wheat disease diagnosis system," Computers and Electronics in Agriculture, vol. 142. Elsevier BV, pp. 369–379, Nov. 2017. doi: 10. 1016/j. compag.2017. 09. 012.

[8] A. Fuentes, S. Yoon, S. Kim, and D. Park, "A Robust Deep learning Based Detector for Real time Tomato Plant Diseases and Pests Recog nition," Sensors, vol. 17, no. 9. MDPI AG, p. 2022, Sep. 04, 2017. doi:10. 3390/s17092022.

[9] M. Kerkech, A. Hafiane, and R. Canals, "Deep leaning approach with colorimetric spaces and vegetation indices for vine diseases detection in UAV images," Computers and Electronics in Agriculture, vol. 155. Elsevier BV, pp. 237–243, Dec. 2018. doi: 10. 1016/j. compag. 2018.10. 006.

[10] Y. Adhitya, G. S. Mulyani, M. Ko¨ ppen, and J. S. Leu, "IoT and Deep LearningBased Farmer Safety System," Sensors, vol. 23, no. 6. MDPI AG, p. 2951, Mar. 08, 2023. doi: 10. 3390/s23062951.

[11] G. Hu, H. Wu, Y. Zhang, and M. Wan, "A low shot learning method for tea leaf's disease identification," Computers and Electronics in Agriculture, vol. 163. Elsevier BV, p. 104852, Aug. 2019. doi: 10. 1016/j. compag. 2019. 104852.

[12] S. Coulibaly, B. KamsuFoguem, D. Kamissoko, and D. Traore, "Deep neural networks with transfer learning in millet crop images," Computers in Industry, vol. 108. Elsevier BV, pp. 115–120, Jun. 2019. doi: 10.1016/j. compind. 2019. 02. 003.

[13] R. Rathore and S. Pratap Singh Rathore, "Machine Learning Applications in Human Resource Management: Predicting Employee Turnover and Performance," International Journal for Global Academic & Scientific Research, vol. 3, no. 2. International Consortium of Academic Professionals for Scientific Research, pp. 48–59, Jul. 02, 2024. doi: 10.55938/ijgasr. v3i2. 77.

[14] A. Picon, A. Alvarez-Gila, M. Seitz, A. Ortiz-Barredo, J. Echazarra, and A. Johannes, "Deep convolutional neural networks for mobile capture device-based crop disease classification in the wild," Computers and Electronics in Agriculture, vol. 161. Elsevier BV, pp. 280–290, Jun.2019. doi: 10. 1016/j. compag. 2018. 04. 002.

[15] M. Rahnemoonfar and C. Sheppard, "Deep Count: Fruit Counting Based on Deep Simulated Learning," Sensors, vol. 17, no. 4. MDPI AG, p. 905, Apr. 20, 2017. doi: 10. 3390/s17040905.

[16] B. Veeramani, J. W. Raymond, and P. Chanda, "DeepSort: deep convolu tional networks for sorting haploid maize seeds," BMC Bioinformatics, vol. 19, no. S9. Springer Science and Business Media LLC, Aug. 2018. doi: 10. 1186/s12859-018-2267-2.

[17] P. Kaushik, S. P. S. Rathore, L. Sachdeva, M. Poonia, D. Singh, and L. Bir, "Intelligent transportation systems trusted user's security and privacy," in 2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), 2024.

[18] Y. Altuntas¸, Z. Co¨ mert, and A. F. Kocamaz, "Identification of haploid and diploid maize seeds using convolutional neural networks and a transfer learning approach," Computers and Electronics in Agriculture, vol. 163. Elsevier BV, p. 104874, Aug. 2019. doi: 10. 1016/j. compag.2019. 104874.

Authorized licensed use limited to: Nitte Meenakshi Institute of Technology. Downloaded on August 13,2025 at 05:21:46 UTC from IEEE Xplore. Restrictions apply.